

PHARMACOVIGILANCE- TASK 2

CASE ANALYSIS OF ADR IDENTIFICATION AND TIMELINE CORRELATION

CLINICAL RECORD LINKAGE ANALYSIS: ADR-9001 to ADR 9008

1. OBJECTIVE: To identify the suspected drug and correlate the timeline of the reported adverse event with drug exposure using the given clinical linkage record and standardized pharmacological criteria.

2. ASSESSMENT APPROACH:

- Identifying the suspected drug and clinical indication.
- Correlating the reported time to onset with drug exposure.
- Assessing the clinical and pharmacological compatibility between the suspected drug and reported event using standardized pharmacological criteria.
- Documenting the severity and outcome as recorded.

3. CLINICAL LINKAGE ANALYSIS OF RECORDED CASES:

ADR-9001-AMOXICILLIN

patient	45-year-old female	Indication	Bacterial infection
Suspected drug	Amoxicillin	Adverse event	Skin rash and hives
Time to onset	2days	Severity	mild
outcome	Recovered	Reporter	pharmacist

Case ADR-9001 describes a 45-year-old female receiving amoxicillin for a bacterial infection. She developed a skin rash and hives 2 days after exposure. The event was recorded as mild, and the patient subsequently recovered. The reported clinical presentation is compatible with a cutaneous hypersensitivity reaction associated with penicillin-class antibiotics. The short time interval between drug exposure and onset provides a “plausible” temporal relationship supporting the suspected association. However, the available record does not provide information regarding concomitant medicines, alternative causes, dechallenge, or rechallenge. Therefore, based on the available information, the WHO-UMC causality assessment is classified as Possible. The reaction is classified as a Type B(BIZZARE) reaction, as it represents an unpredictable immune-mediated response rather than a direct predictable pharmacological effect.

ADR-9002-ATORVASTATIN

patient	62-year-old male	Indication	Hyperlipidemia
----------------	------------------	-------------------	----------------

Suspected drug	Atorvastatin	Adverse event	Severe myalgia
Time to onset	14 days	Severity	moderate
outcome	Recovering	Reporter	Physician

Case ADR-9002 describes a 62-year-old male receiving atorvastatin for hyperlipidemia. He developed severe myalgia 14 days after initiation of treatment. The event was recorded as moderate in severity, and the patient was recovering. Myalgia is consistent with the recognised muscle-related adverse-effect profile of statin therapy. The 14-day interval provides a plausible temporal relationship, while the known association between atorvastatin and muscle symptoms provides pharmacological plausibility. Nevertheless, the available case information does not include other possible causes of muscle pain, concomitant medicines, laboratory findings, dechallenge, or rechallenge information. Therefore, the WHO-UMC causality assessment is Possible. The reaction is classified as a Type A (Augmented) reaction, as it is related to the known pharmacological effects of statin therapy.

ADR-9003-CIPROFLOXACIN

patient	71-year-old female	Indication	UTI
Suspected drug	ciprofloxacin	Adverse event	Tendinitis
Time to onset	7days	Severity	severe
outcome	hospitalized	Reporter	Physician

Case ADR-9003 describes a 71-year-old female receiving ciprofloxacin for a urinary tract infection. She developed tendinitis 7 days after exposure, and the reaction was classified as severe and resulted in hospitalisation. Tendinitis is a recognised adverse reaction associated with fluoroquinolone therapy. The onset within 7 days provides a plausible temporal relationship, while the known association between ciprofloxacin and tendon-related toxicity provides pharmacological plausibility. However, information regarding other risk factors, concomitant medicines, alternative causes, and dechallenge or rechallenge is not available. Consequently, the WHO-UMC causality assessment is Possible. The reaction is classified as a Type B (Bizarre) reaction, as it represents an unpredictable reaction that is not a direct extension of the drug's intended therapeutic pharmacological action.

ADR-9004-IBUPROFEN

patient	28-year-old male	Indication	headache
Suspected drug	ibuprofen	Adverse event	Gastric ulcer bleed
Time to onset	5days	Severity	severe
outcome	Hospitalized	Reporter	patient

Case ADR-9004 describes a 28-year-old male who received ibuprofen for headache and subsequently developed gastric ulcer bleeding 5 days after exposure. The reaction was classified as severe and required hospitalization. The gastrointestinal reaction is pharmacologically plausible and consistent with the known gastrointestinal effects of NSAID therapy. The 5-day interval provides a plausible temporal relationship between drug exposure and the reported event. However, the case record does not provide information about previous gastrointestinal disease, concomitant medicines, other bleeding risk factors, dechallenge, or

alternative causes. Therefore, the WHO-UMC causality assessment is Possible. The reaction is classified as a Type A (Augmented) reaction, as it is consistent with the known pharmacological effects of NSAID therapy.

ADR-9005-LISINOPRIL

patient	53-year-old female	Indication	hypertension
Suspected drug	lisinopril	Adverse event	Persistent dry cough
Time to onset	21 days	Severity	mild
outcome	recovered	Reporter	pharmacist

Case ADR-9005 describes a 53-year-old female receiving lisinopril for hypertension. She developed persistent dry cough 21 days after exposure. The reaction was classified as mild, and the patient recovered. Dry cough is a well-established adverse effect associated with ACE-inhibitor therapy. The 21-day interval provides a plausible temporal relationship, while the known association between lisinopril and persistent dry cough provides pharmacological plausibility. The available information does not indicate whether other causes of cough were excluded or whether symptoms resolved following withdrawal of the medicine. Therefore, the WHO-UMC causality assessment is Possible. The reaction is classified as a Type A (Augmented) reaction, as it is associated with the known pharmacological effects of ACE inhibition.

ADR-9006-METFORMIN

patient	39-year-old male	Indication	Diabetes
Suspected drug	metformin	Adverse event	Nausea and diarrhoea
Time to onset	1 day	Severity	mild
outcome	recovered	Reporter	patient

Case ADR-9006 describes a 39-year-old male receiving metformin for diabetes who developed nausea and diarrhoea 1 day after exposure. The reaction was mild and the patient recovered. Nausea and diarrhoea are recognized gastrointestinal adverse effects associated with metformin, particularly around treatment initiation. The very short interval of one day provides a plausible temporal relationship, and the clinical presentation is pharmacologically plausible based on the known gastrointestinal effects of metformin. However, the record does not provide information regarding the patient's baseline gastrointestinal condition, concomitant medicines, alternative causes, or response following withdrawal. Therefore, the WHO-UMC causality assessment is Possible. The reaction is classified as a Type A (Augmented) reaction, as it is consistent with the known pharmacological effects of metformin therapy.

ADR-9007-WARFARIN

patient	60-year-old male	Indication	Atrial fibrillation
----------------	------------------	-------------------	---------------------

Suspected drug	Warfarin	Adverse event	Uncontrolled hemorrhage
Time to onset	10 days	Severity	Life-threatening
outcome	hospitalized	Reporter	Physician

Case ADR-9007 describes a 60-year-old male receiving warfarin for atrial fibrillation who developed uncontrolled hemorrhage 10 days after exposure. The event was classified as life-threatening and resulted in hospitalization. Excessive anticoagulant effect and bleeding are pharmacologically plausible consequences of warfarin therapy. The occurrence of hemorrhage 10 days after exposure provides a plausible temporal relationship and is clinically compatible with the pharmacological action of warfarin. However, the provided record does not include INR values, warfarin dose, interacting medicines, underlying bleeding risks, or dechallenge information. Therefore, the WHO-UMC causality assessment is Possible. The reaction is classified as a Type A (Augmented) reaction, as excessive bleeding is consistent with the known pharmacological effects of anticoagulation

ADR-9008-PACLITAXEL

patient	50-year-old female	Indication	chemotherapy
Suspected drug	Paclitaxel	Adverse event	Peripheral neuropathy
Time to onset	30 days	Severity	moderate
outcome	Not recovered	Reporter	oncologist

Case ADR-9008 describes a 50-year-old female receiving paclitaxel as part of chemotherapy who developed peripheral neuropathy 30 days after exposure. The reaction was classified as moderate, and the patient had not recovered at the time of reporting. Peripheral neuropathy is a recognised adverse effect associated with taxane therapy. The 30-day interval provides a plausible temporal relationship, while the known association between paclitaxel and peripheral neuropathy provides pharmacological plausibility. However, the available record does not provide information regarding cumulative dose, concomitant neurotoxic medicines, alternative causes, dechallenge/rechallenge. Therefore, the WHO-UMC causality assessment is Possible. The reaction is classified as a Type A (Augmented) reaction, as it is consistent with the known toxicity profile of paclitaxel.

4.CONCLUSION:

The eight cases demonstrate clinically plausible relationships between the suspected medicine and reported adverse reactions. Based on the given information they are classified as possible according to WHO-UMC criteria with reactions as TYPE A (AUGUMENTED) or TYPE B (BIZZARE).